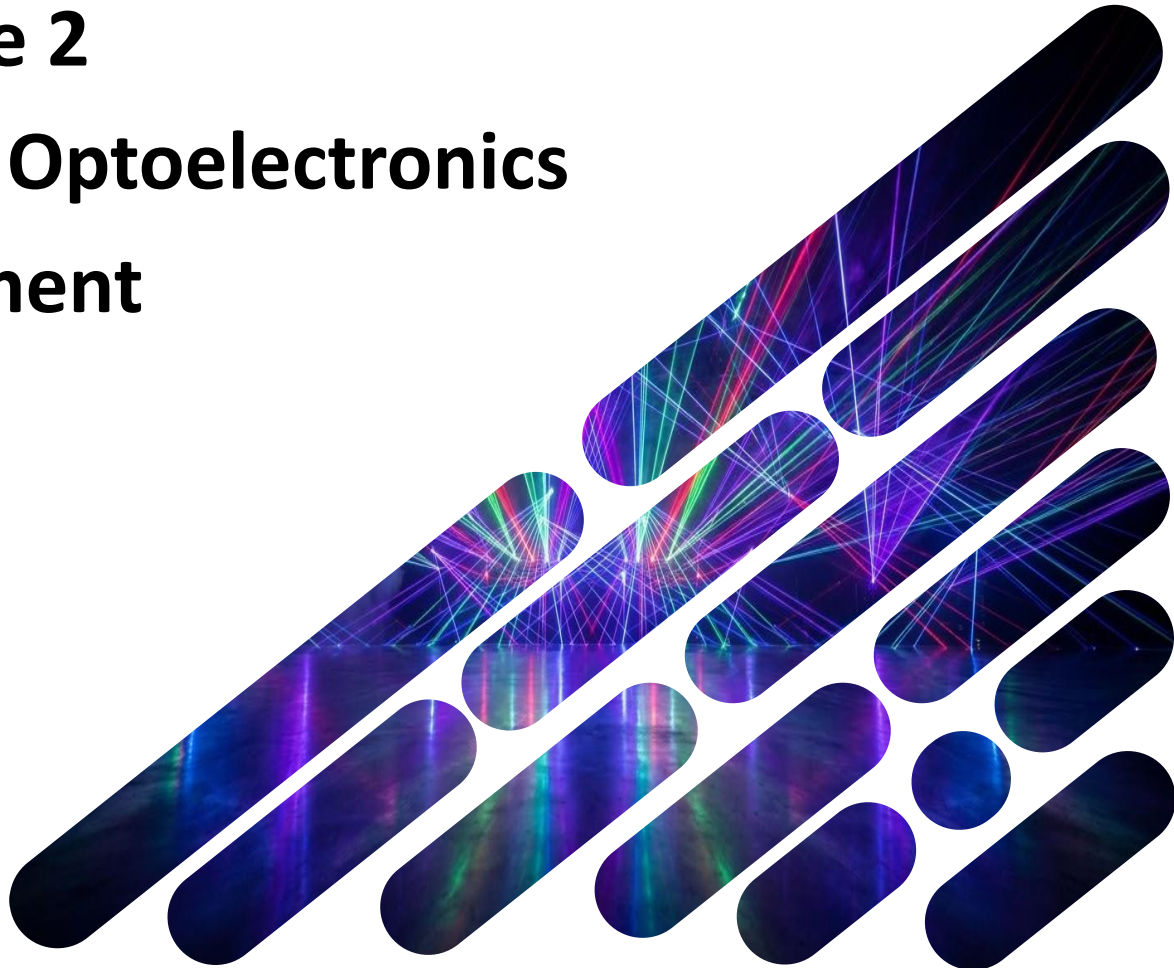
Nonlinear Optics

Lecture 2

Laser Engineering & Optoelectronics Department



Lect. Sujood Basim Jabr

Introduction

Nonlinear optical susceptibility is one of the fundamental properties used to describe the response of materials to intense optical electric fields, such as those produced by high-intensity lasers.

When light propagates through a material, the optical electric field interacts with the electrons of atoms and molecules. At sufficiently high field strengths, the response of the material may no longer be directly proportional to the electric field. As a result, new optical phenomena can occur.

Definition of Nonlinear Optical Susceptibility

Nonlinear optical susceptibility is a material property that describes how a medium responds to an optical electric field in a nonlinear manner.

The first-order susceptibility is denoted by $\chi^{(1)}$,the second-order susceptibility by $\chi^{(2)}$,and the third-order susceptibility by $\chi^{(3)}$.

These quantities determine the nature and strength of different nonlinear optical effects that can occur in a material.

The Electric Polarization Equation

The total electric polarization of a material can be expressed as an expansion in powers of the electric field:

$$P=\varepsilon 0\ \chi^{(1)}\ E+\varepsilon 0\ \chi^{(2)}\ E2+\varepsilon 0\chi^{(3)}E3+\cdots$$

Where:

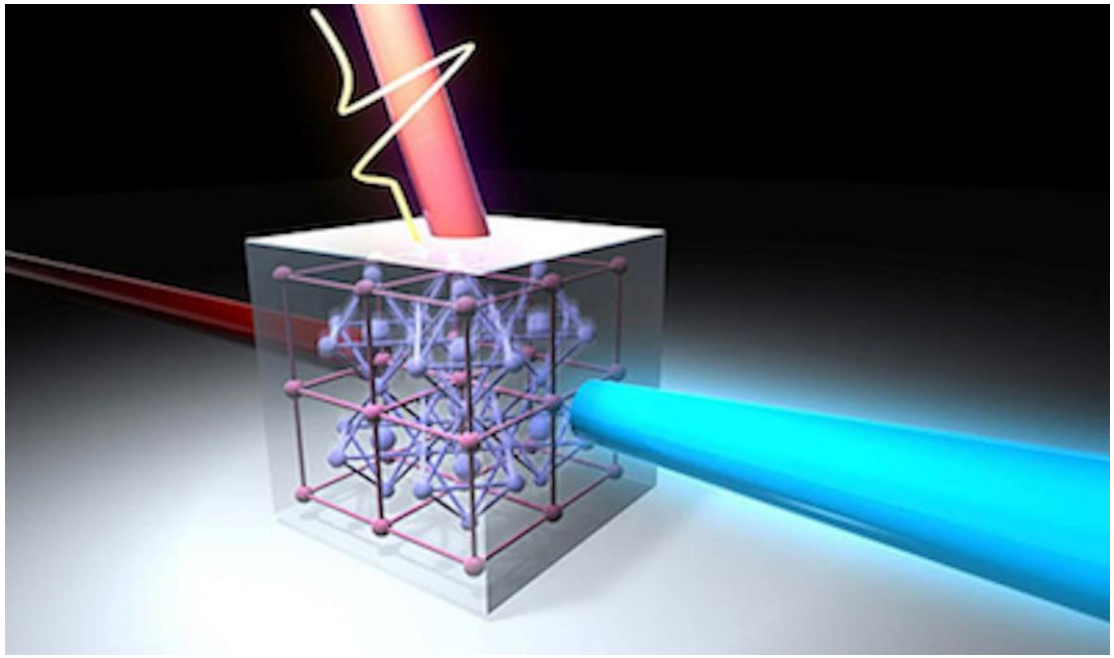
- P: Electric polarization, measured in $\mathrm{C/m^2}$.
- ε_0: Permittivity of free space.
- E: Optical electric field, measured in $\mathrm{V/m}$.
- $\chi^{(1)}$: Linear susceptibility.
- $\chi^{(2)}$: Second-order nonlinear susceptibility.
- $\chi^{(3)}$: Third-order nonlinear susceptibility.

Types of Optical Susceptibility

1.First-Order Susceptibility $X^{(1)}$

First-order susceptibility describes the linear response of a material. It is related to optical properties such as refractive index and absorption in a linear medium.

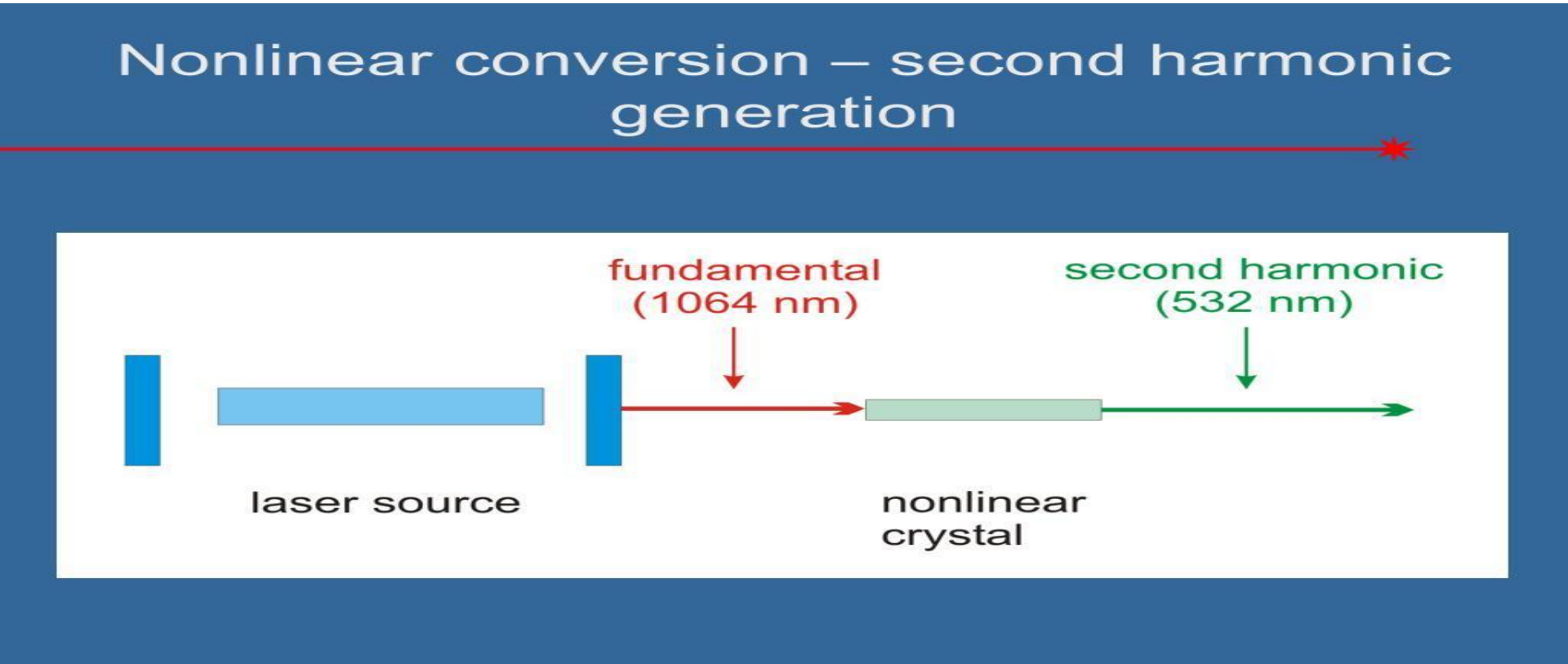
$P^{(1)}=\varepsilon 0\ X^{(1)}\ E$



2. Second-Order Susceptibility $X^{(2)}$

Second-order susceptibility is responsible for effects such as second-harmonic generation, in which light at frequency **f** can generate light at frequency **2f**.

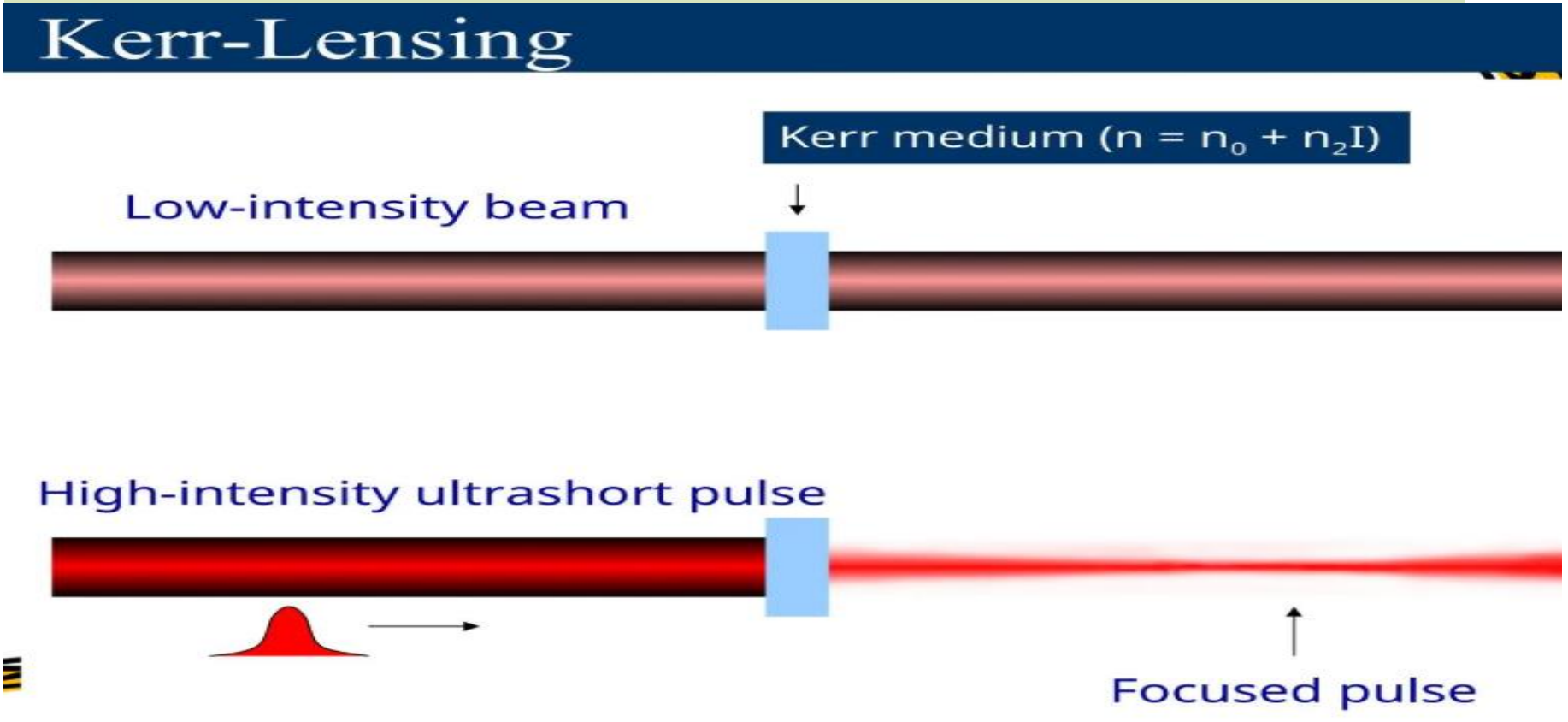
$$P^{(2)}=\varepsilon 0\ \mathbf{X^{(2)}}\ E2$$



3. Third-Order Susceptibility $X^{(3)}$

Third-order susceptibility is associated with effects such as the optical Kerr effect, self-phase modulation, and certain frequency-generation processes.

$P^{(3)}=\varepsilon 0 X^{(3)} E3$



Units of Optical Susceptibility

In the International System of Units (SI), using the polarization expansion given above:

Susceptibility	SI unit
$\chi^{(1)}$	Dimensionless
$\chi^{(2)}$	m/V
$\chi^{(3)}$	$m^2 \setminus V^2$

These units correspond to electric polarization measured in C/m^2 and electric field measured in V/m.

What is the difference between second-order and third-order susceptibility?



Factors Affecting Nonlinear Optical Susceptibility

Several factors influence the nonlinear optical response of a material:

Electric field strength: The nonlinear response becomes more significant as the optical field increases.

Material composition: Different materials have different electronic structures and nonlinear responses.

Crystal symmetry: Symmetry determines whether certain nonlinear optical effects are allowed.

Optical frequency or wavelength: Susceptibility may vary with the frequency of the incident light.

Temperature: Temperature can affect material properties and nonlinear response, depending on the medium and experimental conditions.

Problem:

A material has a second-order nonlinear $X^{(2)}$=2×10^12m/V, The applied electric field is E=1×10^6 V/m .Calculate the second-order nonlinear polarization, given that ε0=8.854× 10^-12 F/m

$$P^{(2)} = \varepsilon_0 \chi^{(2)} E^2$$

$$P^{(2)} = (8.854 \times 10^{-12}) \times (2 \times 10^{-12}) \times (10^6)^2$$

$$\boxed{P^{(2)} = 1.7708 \times 10^{-11}\ \mathrm{C/m^2}}$$

H.W:

A material has a Third-order nonlinear $X^{(3)}$=3×10^-20 m/V, The applied electric field is E=2×10^6 V/m .Calculate the Third-order nonlinear polarization, given that $\varepsilon 0$=8.854× 10^-12 F/m

